

CNS2025: Homework 13

Due: 2025-12-03 23:59

Load the frame-based spike-train data from [hw13-data.npz](#). Use the `spike_train` data as spike indicators (1 for having spike in the time frame, 0 otherwise). The time between frames is given by `delta_t` in seconds in the data file. Use sequence length T_s ranging from 5 to 100 ms (in 5 ms increments) to calculate the *entropy rate* of the spike train. You may see [code13.ipynb](#) for how the entropy rate can be calculated at a given T_s value.

Plot the entropy rate (per second) as a function of $1/T_s$ similar to Fig 4.7 in [slide 4](#) of the lecture.

Extrapolate the entropy rate to the $1/T_s = 0$ Hz limit. What is your estimate of the entropy rate at this limit?

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